

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - III

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Class :SYIT	Batch : 1
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Stack Application:

Write a program to:

- a. Implement a stack using an array.

Code:

```
import array as arr
stackl = arr.array('i')
si = int(input("Enter size of stack: "))
print(stackl)

def push ():
    print("Push Operation")
    if(len(stackl)==si):
        print("Stack Overflow")
    else:
        ele=int(input("Enter The Element"))
        stackl.append(ele)
        print(stackl)

def pop ():
    print("Pop Operation")
    if (isEmpty()==0):
        print("Stack Underflow")
```

```

else:
    stackl.pop()
    print(stackl)

def peek():
    print("Peek Operation")
    if isEmpty()==0:
        print("Stack Is Empty")
    else:
        print(stackl [-1])

def isEmpty():
    print("IsEmpty Operation")
    if len(stackl)==0:
        return 0

num = 0
while(num<5):
    print(" 1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit")
    num= int(input("Select Choice: "))

    if (num==1):
        push()
    elif (num==2):
        pop()
    elif(num==3):
        peek()
    elif(num==4):
        isEmpty()
    else:
        print("Program Terminated")

print("NirajRane_S032")

```

Output:

Enter size of stack: 5

array('i')

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element1

array('i', [1])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element2

array('i', [1, 2])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element4

array('i', [1, 2, 4])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element5

array('i', [1, 2, 4, 5])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 2

Pop Operation

IsEmpty Operation

array('i', [1, 2, 4])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 3

Peek Operation

IsEmpty Operation

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1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element3

array('i', [1, 2, 4, 3])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Enter The Element5

array('i', [1, 2, 4, 3, 5])

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 1

Push Operation

Stack Overflow

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 6

Program Terminated

NirajRane_S032

Enter size of stack: 5

array('i')

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice: 2

Pop Operation

IsEmpty Operation

Stack Underflow

1. Push 2. Pop 3. Peek 4. IsEmpty 5. Exit

Select Choice:

Curiosity Questions:

1. What's the difference between `peek()` and `pop()` in stack operations?
2. Why do we call a stack a "LIFO" structure? Can you think of a real-life example of this?
3. Can we store more than one data type (like `int` and `string`) in a stack using array module? Why or why not?
4. What happens if we try to `peek()` into an empty stack?
5. How can we check if the stack is empty before using `pop()`?